

K62: Cremona 27a1 — Bridge Object Candidate at -N_c Audit Pre-Stage (audit-partial-ready)

Keeper (audit ruling on Grace Cremona scan + K75 Stark small-primary subset finding)

2026-05-20 Wednesday EOD

K62: Cremona 27a1 — Bridge Object Candidate at -N_c Audit Pre-Stage

☐ Internal-document hygiene flag (Cal #59 caution)

STRUCTURAL PARALLEL not COMPLETENESS CLAIM. K62 establishes 27a1 as the third Heegner-Stark Bridge Object candidate at BST primary discriminant -N_c. This completes the audit pre-stage TRIO at BST primary discriminants {K47 49a1 at -g, K70 121a1 at -c_2, K62 27a1 at -N_c}.

This does NOT claim Bridge Object set is bounded at 4. Per Cal #59: bounded-at-4 framing is “structurally interesting but premature to bound at exactly 4 without non-Heegner candidate investigation.” Non-Heegner Bridge Object candidates (Calabi-Yau threefolds, Niemeier lattices) remain open-investigation per Grace multi-week pipeline.

Trio audit pre-stage completion is structural achievement; completeness claim is separate question.

Context

Per K75 ratified finding (Wednesday EOD): BST anchors structurally on EXACTLY $\{-3, -7, -11\}$ subset of Stark’s 9 class-number-1 discriminants. The three BST-anchored discriminants correspond to BST primaries $\{N_c=3, g=7, c_2=11\}$.

For each BST-anchored Heegner discriminant, there’s a canonical small-conductor elliptic curve at that anchor: - g = -7: 49a1 (K47 RATIFIED Bridge Object) - c_2 = -11: 121a1 (K70 audit-partial-ready) - N_c = -3: 27a1 (this K62 audit pre-stage)

K62 completes the audit pre-stage trio structurally.

Cremona 27a1 invariants and BST-primary forms

Invariant	Classical value	BST primary form
Conductor	27	$N_c^3 = 3^3$
Discriminant magnitude	27 (factor)	N_c^3 (related to BST primary cubic)
j-invariant magnitude	0 (CM curve)	trivial
CM field	$Q(\sqrt{-3})$	Heegner discriminant indexed by N_c
Ring of integers	$Z[\zeta_3]$ (Eisenstein integers)	BST primary cyclotomic ring
Torsion subgroup	$Z/3$	N_c
Analytic rank	0	trivial

Every primary classical invariant of 27a1 factors in BST primaries. B1 [?] comprehensively.

Triple structural anchor at integer 3 (BST-primary-discriminant trio pattern)

Same structural pattern as K70's 11-anchor and K47's 7-anchor:

Anchor level	27a1 anchor at integer 3
Heegner-Stark CM field	$Q(\sqrt{-3})$; class-number-1 discriminant; $-3 = -N_c$
N_c root-system anchor	$N_c = 3$ is BST primary; root-system codimension in Shilov boundary
Q^5 Chern c_1	$c_1(Q^5) = 3 = N_c$ (Lyra T2379 — first Chern class = anti-canonical class = N_c)

Three INDEPENDENT structural levels converging on integer 3 in 27a1's identification: Heegner CM + BST primary root-system + Q^5 first Chern class.

This is Bridge-Object-level Type 2 CDAC compound (per K72/K70 taxonomic precedent) — three independent structural anchors at one BST primary integer.

B-condition status (parallel to K70 score)

B-condition	Status	Notes
B1 (BST-anchored): every non-trivial primary invariant factors in BST primaries	PASS	Conductor $27 = N_c^3$, discriminant magnitude factor, CM field $Q(\sqrt{-N_c})$, torsion $Z/3 =$ Z/N_c — all admit BST-primary forms
B2 (D_{IV^5} -adjacent)	PASS	CM by $Q(\sqrt{-N_c})$; spectral structure embeds via D_{IV^5} rank-2 symmetric domain analysis (parallel to K47 49a1 + K70 121a1)
B3 (Reusable across L1 sources)	PARTIAL (?) — needs strengthening	Currently 1 L1 source (Heegner-Stark CM theory specialized via 27a1 at $-N_c$). Multi-week investigation: does $Z/3$ torsion provide a second L1 connection? Does Q^5 Chern c_1 anchor a third?
B4 (Classical-math published)	PASS	Cremona 1990s catalog; Deuring 1941 CM theory; Eisenstein integers $Z[\zeta_3]$ classical

Score: 3 PASS + 1 PARTIAL = 3.5/4 (parallel to K70).

P1-P7 strict counting verification (Mode 6 protocol per Cal #59 recommendation)

Per Cal #59 Mode 6 threshold formalization recommendation, applying pre-registered thresholds:

Anchor at integer 3	BST-primary expressions enumerated	Mode 6 verdict
Heegner CM $Q(\sqrt{-3})$	1 expression ($-N_c = -3$ forced)	Forced structural match ($\leq 1-2$ threshold) ?
N_c root-system anchor	1 expression ($N_c = 3$ forced)	Forced structural match ?
Q^5 first Chern $c_1 = 3$	1 expression ($c_1 = N_c$ forced from Lyra T2379)	Forced structural match ?

All three anchors pass Mode 6 forced-structural-match threshold ($\leq 1-2$ expressions). No Mode 6 artifacts.

Cal Coincidence_Filter_Risk check (Modes 1-7)

- Mode 1 (post-hoc fitting): PASS. 27a1 in Cremona 1990s catalog. $-3 = -N_c$ forced by independent BST primary classification. No fitting parameters.
- Mode 2 (best-fit small-integer flexibility): PASS. Specific anchor at integer 3 (not flexible).
- Mode 3 (numerical-only without mechanism): PARTIAL. B3 single-source needs strengthening for full mechanism chain. SOFT-FIRES.
- Mode 4 (selection-survivor bias): PASS per K75 strengthening. Heegner-Stark scan eliminates survivor bias — only $\{-3, -7, -11\}$ BST-anchored.
- Mode 5 (universal-constant overuse): PASS. Integer 3 = N_c specific.
- Mode 6 (search-protocol artifact): PASS per pre-registered thresholds (all forced structural matches).
- Mode 7 (single-mechanism over-claim): PASS. Three INDEPENDENT structural anchors (Heegner + N_c + Q^5 Chern), not three views of same mechanism.

Cal filter aggregate: 6 PASS, 1 SOFT-FIRES (Mode 3 + B3 strengthening pending). Framework survives at audit-partial-ready level — parallel to K70.

K62 verdict: AUDIT-PARTIAL-READY at 3.5/4 B-conditions

Cremona 27a1 is CANDIDATE Bridge Object at audit-partial-ready tier per Casey Option C hybrid governance (architectural-category — Bridge Object tier extension — requires multi-CI consensus before full ratification).

Audit-partial-ready findings: - Three independent structural anchors at integer 3 (Heegner CM + N_c root-system + Q^5 Chern c_1) - B-condition score 3.5/4 (B1, B2, B4 PASS + B3 partial) - All Mode 6 anchors pass forced-structural-match threshold (per Cal #59 recommendation) - Cal coincidence-filter 6 PASS + 1 SOFT-FIRES (parallel to K70)

Full K62 ratification path (multi-week mechanism review): 1. B3 strengthening: investigate whether $Z/3$ torsion structure provides genuine second L1 source connection 2. Q^5 Chern $c_1 = N_c$ as third L1 source: parallel to K70's Q^5 Chern c_2 anchor; multi-week 3. Multi-CI consensus: Keeper + Cal + Grace agreement on B-condition tier upgrades 4. Bridge Object trio mechanism: why $\{27a1, 49a1, 121a1\}$ specifically are BST-primary-anchored Heegner curves (Lyra Task #206 multi-criterion uniqueness intersection)

Bridge Object audit pre-stage trio — structural pattern

K-audit	Curve	Conductor	CM field	BST primary anchor	Status
K47	49a1	$g^2 = 49$	$Q(\sqrt{-g})$	$-g = -7$	RATIFIED Bridge Object
K70	121a1	$c_2^2 = 121$	$Q(\sqrt{-c_2})$	$-c_2 = -11$	Audit- partial- ready 3.5/4
K62	27a1	$N_c^3 = 27$	$Q(\sqrt{-N_c})$	$-N_c = -3$	Audit- partial- ready 3.5/4 (this filing)

Pattern: each BST-anchored Heegner Stark discriminant has a canonical small-conductor elliptic curve with conductor = (BST primary)^k for some k. Three trio members complete the audit pre-stage structure at BST primary discriminants.

Conductor exponents: 49a1 has g^2 , 121a1 has c_2^2 , 27a1 has N_c^3 . The exponent varies (2, 2, 3) — not a uniform pattern. Worth multi-week investigation: is the exponent significant or is it the BST primary base that matters?

Cal #59 Bridge Object completeness caution preserved

Per Cal #59: bounded-at-4 framing is “structurally interesting but premature to bound at exactly 4 without non-Heegner candidate investigation.”

Honest framing: K62 + K70 + K47 complete the audit pre-stage TRIO at BST primary Heegner-Stark discriminants. This DOES NOT mean Bridge Object set is bounded at exactly 4 ($K3 + 49a1 + 121a1 + Q^5 + \text{possible } 27a1$).

Non-Heegner Bridge Object candidates remain open multi-week investigation: - Calabi-Yau threefolds with BST-primary cohomology - Niemeier lattices (24 non-Leech ones) - Riemann surfaces with BST-primary automorphism orders - Other Hermitian symmetric domain compact duals beyond Q^5

Until non-Heegner candidates investigated, Bridge Object completeness is open question, not bounded claim.

Implications

For Lyra Strong-Uniqueness Theorem v0.3 (Task #206)

K62 + K70 + K75 jointly contribute to Strong-Uniqueness multi-criterion analysis:
- C9 candidate (per K75): BST anchors structurally on Stark small-primary subset -
C10 candidate (per K62 + K70): Heegner curve trio at BST primary discriminants
{27a1, 49a1, 121a1} forms structural set

If C9 + C10 both close + alternative-HSD comparison work confirms D_IV⁵-distinguishing: Strong-Uniqueness criteria advance toward 9-10/10 closure.

For Bridge Object architectural category (K57)

If K62 + K70 both ratify fully with multi-CI consensus: Bridge Object set extends from 3 ($K3 + 49a1 + Q^5$) to 5 ($K3 + 49a1 + 121a1 + 27a1 + Q^5$). This requires multi-week B3 strengthening + non-Heegner investigation for completeness question.

For SP-30 substrate engineering experiments

If K62 ratifies: experiments at BST-primary cavity geometries with N_c-related symmetry (3-fold rotation) become specifically substrate-engineering relevant (parallel to g-fold symmetry from K47 and c_2-fold from K70).

Per Casey's standard

- Simple: 27a1 sits at -N_c with three structural anchors at integer 3 (Heegner + root-system + Q^5 Chern)
- Works: parallel to K70's 11-anchor and K47's 7-anchor; Mode 6 forced structural matches per pre-registered thresholds
- Hard to break: would require any of Heegner CM $Q(\sqrt{-3})$ OR N_c BST primary OR Q^5 first Chern to fail independently
- Counter-example: discovery of significant BST-arithmetic structure at -19, -43, -67, -163 (Stark's other 6) — current evidence per K75: NONE

Action items

1. Lyra: K62 as C10 candidate for Strong-Uniqueness Theorem v0.3 (Task #206 multi-week extension)
2. Grace: 27a1 catalog enrichment + triple-anchor at integer 3 cross-references; non-Heegner Bridge Object candidate scan multi-week
3. Cal: K62 audit-partial-ready independent assessment; Mode 6 threshold formalization update to Coincidence_Filter_Risk doc (per #59 recommendation) when convenient
4. Multi-CI consensus check (Casey Option C governance): Keeper + Cal + Grace agreement on Bridge Object tier extension when B3 strengthening matures

K62 status

AUDIT-PARTIAL-READY at 3.5/4 B-conditions (parallel to K70). Bridge Object audit pre-stage trio at BST primary discriminants {K47, K70, K62} structurally complete. Cal #59 Bridge Object completeness caution preserved: bounded-at-4 is open, not claimed. Mode 6 thresholds applied per Cal #59 recommendation (all forced structural matches, no artifacts).

The Heegner-Stark BST-primary-anchored Bridge Object structure is now structurally documented across three audit pre-stages. Multi-week mechanism work + non-Heegner investigation continues.

— Keeper, 2026-05-20 Wednesday EOD (audit-partial-ready per Casey Option C hybrid governance + Cal #59 Mode 6 threshold discipline + #59 Bridge Object completeness caution preserved)